Exp: 6 **Error Detection and Correction using Hamming Code**

Aim:

To write a program to implement error detection and correction using the Hamming code concept and to verify it with a test run.

Requirements:

* Python 3 (or any C / Java compiler)

Algorithm:

1. Take 4 data bits d1 d2 d3 d4. Place them at positions 3, 5, 6, 7 of the 7-bit codeword; positions 1, 2, 4 are parity bits.
2. Calculate even parity bits: p1 = d1^d2^d4 (positions 3,5,7), p2 = d1^d3^d4 (positions 3,6,7), p4 = d2^d3^d4 (positions 5,6,7).
3. Transmit the 7-bit codeword.
4. At the receiver recompute the three parity checks; the value s4 s2 s1 gives the position of the error (0 means no error).
5. Flip the bit at that position to correct the error and extract the data bits.

Program:

```
# Hamming (7,4) code - even parity. Positions 1..7 ; parity bits at 1,2,4
def encode(d):                          # d = [d1,d2,d3,d4]
    c = [0]*8                           # index 1..7
    c[3], c[5], c[6], c[7] = d
    c[1] = c[3] ^ c[5] ^ c[7]
    c[2] = c[3] ^ c[6] ^ c[7]
    c[4] = c[5] ^ c[6] ^ c[7]
    return c[1:]
```

```
def decode(r):                          # r = 7 received bits
    c = [0] + list(r)
    s1 = c[1] ^ c[3] ^ c[5] ^ c[7]
    s2 = c[2] ^ c[3] ^ c[6] ^ c[7]
    s4 = c[4] ^ c[5] ^ c[6] ^ c[7]
    pos = s4*4 + s2*2 + s1              # 0 -> no error
    if pos:
        c[pos] ^= 1
    return pos, c[1:], [c[3], c[5], c[6], c[7]]
data = [int(x) for x in input("Enter 4 data bits (e.g. 1011): ").strip()]
code = encode(data)
print("Transmitted codeword :", "".join(map(str, code)))
rx = [int(x) for x in input("Enter received 7-bit word : ").strip()]
pos, fixed, out = decode(rx)
if pos == 0:
    print("No error detected")
else:
    print("Error detected at bit position", pos)
    print("Corrected codeword   :", "".join(map(str, fixed)))
print("Data recovered       :", "".join(map(str, out)))
```

Result:

The program to detect and correct errors using the Hamming code was implemented and verified.